Lecture 9: Other Applications of Reinforcement Learning

Physics-informed models, molecules, robotics, and scientific frontiers

FYS4411 — Reinforcement Learning and Agentic AI
University of Oslo, Department of Physics

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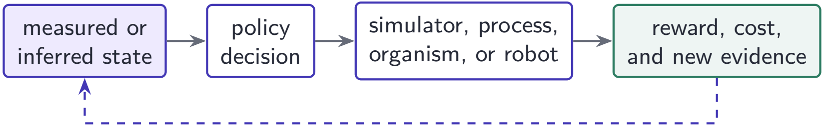
Opening proposition: RL is a method for controlled experimentation

Reinforcement learning is most compelling when a researcher cannot prescribe the best sequence of decisions, but can construct a loop in which decisions produce informative consequences.



The research value is not “trial and error” by itself. It is the ability to optimize decisions under delayed effects, uncertainty, constraints, and feedback.

A scientific problem becomes RL only after decisions are explicit

Element	Scientific interpretation	Example: temperature control
State s_t	information available for decision	temperature field, boundary flux, uncertainty
Action a_t	controllable intervention	heater power or boundary condition
Transition	physical or learned dynamics	heat equation plus disturbances
Reward r_t	operational objective and penalties	tracking error, energy, constraint violation
Policy π	rule for repeated decisions	feedback controller

If there is no meaningful sequence of interventions, supervised learning, optimal design, Bayesian optimization, or conventional numerical methods may be more appropriate.

Why RL can motivate new research

Many scientific systems share four properties:

- actions alter the distribution of future observations;
- the quantity of interest appears only after a long trajectory;
- accurate dynamics are incomplete, expensive, or partially observed;
- constraints and competing objectives make one-step optimization insufficient.

Research opportunity

RL provides a common language for adaptive experiments, active simulation, closed-loop control, and sequential scientific design. The strongest work combines that language with domain structure rather than replacing the domain model.

Learning goals

After this lecture, you should be able to

- formulate scientific and engineering applications as MDPs or POMDPs without hiding important constraints;
- explain how RL can interact with PINNs, PDE solvers, molecular simulation, and classical control;
- design detailed state, action, reward, simulator, and validation choices for representative applications;
- distinguish simulation success, retrospective evaluation, laboratory validation, and real deployment;
- identify research questions where offline RL, model-based RL, safe RL, and multi-objective RL are genuinely useful.

Organization

Part	Central question
I. Research formulation	When does a scientific problem justify RL?
II. PINNs and PDE control	How can physics structure learning and control?
III. Molecular systems	How can policies search chemical and dynamical spaces?
IV. Robotics and control	How are learned policies transferred to physical systems?
V. Wider applications	Where is RL promising, demonstrated, or still speculative?
VI. Research frontier	What should the next generation of scientific RL optimize?

Part I: From research question to decision process

RL is justified by sequential intervention, not by complexity alone

The Markov decision process

An MDP is $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, P, r, \gamma \rangle$. A policy $\pi_\theta(a \mid s)$ is optimized for

$$J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t=0}^{T-1} \gamma^t r(s_t, a_t, s_{t+1}) \right].$$

For research applications, each mathematical object should have a measurable operational interpretation:

- P may be a physical simulator, empirical process, learned world model, or experiment;
- r encodes a scientific or engineering objective but is rarely the objective itself;
- γ and T determine which timescales the policy values;
- a trajectory τ is the primary unit of validation.

Partial observability is the scientific default

Experiments rarely expose the complete physical state. A POMDP adds observations $o_t \sim O(\cdot \mid s_t)$ and a belief state

$$b_t(s) = P(s_t = s \mid o_{0:t}, a_{0:t-1}).$$

latent state s_t → sensor or measurement → filter, history, or recurrent state → action a_t

In molecular dynamics, relevant collective variables may be unknown. In medicine, disease severity is only imperfectly observed. In robotics, velocity, contact, and friction must be inferred from noisy sensors.

RL is adjacent to optimization and control—not a replacement

Method	Best suited to	Central assumption
Optimal control / MPC	known or estimated dynamics	model is adequate over planning horizon
Bayesian optimization	expensive static experiments	each query has no important state transition
Supervised learning	prediction from labeled examples	target is available in data
RL	adaptive sequential intervention	decisions alter future states and evidence

Hybrid methods are often strongest: MPC provides constraints and short-horizon planning; RL learns residuals, objectives, or adaptation strategies.

When should a researcher use RL?

RL is plausible when all of the following are reasonably true:

- a decision must be repeated as the system evolves;
- delayed effects or exploration create value beyond greedy optimization;
- a simulator, logged dataset, or safe experimental interface supports learning;
- objectives and constraints can be measured with useful fidelity;
- validation can compare against strong non-RL baselines.

Warning sign

If the reward is easy to optimize but weakly connected to the scientific goal, RL will efficiently solve the wrong problem.

Scientific control is usually constrained control

Maximizing expected reward is insufficient when violations are unacceptable. A constrained MDP solves

$$\max_{\pi} \; J_r(\pi) \quad \text{subject to} \quad J_{c_i}(\pi) = \mathbb{E}_{\pi}\left[\sum_t \gamma^t c_i(s_t, a_t)\right] \leq d_i.$$

Soft costs

- energy or reagent consumption;
- wear, latency, and compute;
- deviation from a reference.

Hard constraints

- robot torque and collision limits;
- dose and toxicity bounds;
- conservation and stability conditions.

Model-based and offline RL address expensive interaction

Model-based RL

- learn or reuse $\hat{P}(s' \mid s, a)$;
- plan or generate imagined rollouts;
- quantify model uncertainty;
- periodically correct with real evidence.

Offline RL

- learn from a fixed dataset $\mathcal{D}$;
- avoid unrestricted exploration;
- penalize out-of-distribution actions;
- rely on coverage and causal assumptions.

Scientific systems often need both: a physics-based or learned model for planning and a conservative offline objective grounded in historical data.

Evaluation requires counterfactual discipline

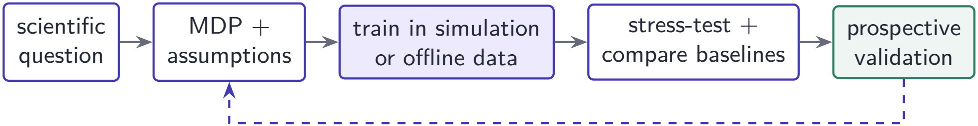
The policy asks what would happen under actions that may differ from those in the data. For trajectories generated by behavior policy μ, importance weighting illustrates the difficulty:

$$\hat{J}(\pi) = \frac{1}{N}\sum_{i=1}^{N}\left(\prod_{t}\frac{\pi(a_t^i \mid s_t^i)}{\mu(a_t^i \mid s_t^i)}\right) G_i.$$

The product can have extreme variance, while hidden confounding may invalidate the estimate entirely.

- Use simulation and off-policy evaluation as filters, not final proof.
- Test temporal and institutional distribution shift.
- Require prospective experiments for consequential claims.

A disciplined scientific RL workflow



1. Define the scientific claim before tuning the reward.
2. Preserve classical, random, greedy, and model-based baselines.
3. Pre-specify success, constraints, uncertainty, and stopping.
4. Validate where the claimed benefit will actually be used.

Part II: PINNs, PDEs, and learned control

Physics can structure both the world model and the objective

A PINN is a physics-constrained function approximator

For a PDE

$$\mathcal{N}[u](x,t) = f(x,t), \qquad \mathcal{B}[u](x,t) = g(x,t),$$

a neural approximation $u_\phi(x,t)$ is trained using automatic differentiation:

$$\mathcal{L}_{\mathrm{PINN}}(\phi) = \lambda_r \frac{1}{N_r} \sum_{i=1}^{N_r} \left|\mathcal{N}[u_\phi](x_i,t_i) - f_i\right|^2 + \lambda_b \mathcal{L}_{\mathrm{boundary}} + \lambda_d \mathcal{L}_{\mathrm{data}}.$$

PINN training is ordinarily gradient-based supervised/self-supervised optimization, not reinforcement learning. RL enters when observations or controls must be selected sequentially.

Motivated by Raissi, Perdikaris, and Karniadakis, *Physics-informed neural networks* (JCP, 2019).

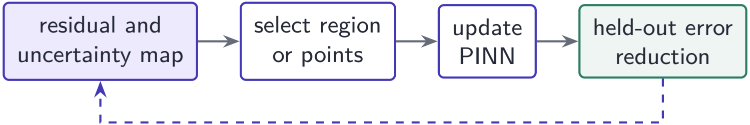
Four legitimate interfaces between PINNs and RL

Interface	RL decision	PINN role
Adaptive collocation	where to sample PDE residual next	trainable surrogate
Active sensing	which sensor, time, or location to query	state reconstruction
Model-based control	which intervention to apply	differentiable dynamics model
Physics-informed policy	how to act under constraints	residual or conservation penalty

The coupling is useful only when it reduces expensive solver calls, improves partial-state estimation, or enforces physical structure better than a simpler method.

Adaptive collocation as a sequential decision problem

Uniform collocation wastes capacity in easy regions and may miss shocks, layers, or rapidly changing residuals.



One formulation uses state s_k = residual histogram, uncertainty, and sampling history; action a_k = allocation of a collocation budget; reward = validation improvement per new point.

Residual-based adaptive sampling is already a strong non-RL baseline. RL is justified only if the allocation decision has delayed or path-dependent effects.

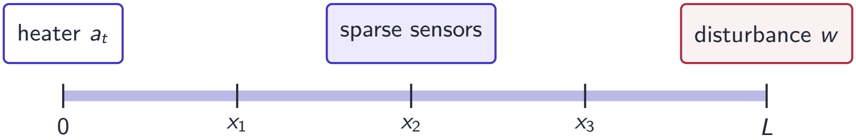
Motivated by Wu et al., adaptive sampling for PINNs (2023); Subramanian et al. (2022).

Detailed example: boundary control of a heated rod

Consider

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2} + b(x)a_t + w(x,t), \qquad x \in [0, L],$$

with initial condition $u(x,0) = u_0(x)$ and controlled boundary or distributed heating a_t.



Goal: track a spatial reference $u^\star(x,t)$ while limiting energy, gradients, and maximum temperature.

The heat-control MDP

State	sensor values, recent controls, PINN-estimated field $\hat{u}_\phi(x, t)$, uncertainty
Action	heater power $a_t \in [a_{\min}, a_{\max}]$ or boundary temperature
Transition	high-fidelity PDE solver during evaluation; PINN surrogate during some training
Reward	tracking, energy, smoothness, and safety penalties
Episode	a heating cycle with randomized initial conditions and disturbances

The PINN converts sparse measurements into a field estimate and can accelerate rollouts. The policy must still be validated against a numerical solver and physical experiment.

Reward and constraints must reflect the PDE objective

A possible step reward is

$$r_t = -\underbrace{\|u_t - u_t^\star\|_{L^2(0,L)}^2}_{\text{tracking}} - \lambda_a a_t^2 - \lambda_\Delta (a_t - a_{t-1})^2 - \lambda_T \int_0^L [\max(0, u_t - u_{\max})]^2 dx.$$

For a hard temperature constraint, use a safety filter rather than relying only on λ_T:

$$a_t^{\mathrm{safe}} = arg \min_{a \in \mathcal{A}} \|a - a_t^{\mathrm{RL}}\|^2 \quad \text{s.t.} \quad \hat{u}_{t+1}(x; a) \le u_{\max} - \delta.$$

The margin δ should incorporate surrogate and sensor uncertainty.

Code sketch: train with a surrogate, verify with the solver

```
for episode in range(num_episodes):
    params = sample_physics_and_disturbances()
    obs = reset_system(params)

    for t in range(horizon):
        field, sigma = pinn.reconstruct(obs, action_history)
        proposed = policy.sample(field, sigma)
        action = safety_filter(proposed, field, sigma)

        # Cheap transitions during training; scheduled truth checks.
        next_obs = surrogate.step(field, action, params)
        if t % truth_interval == 0:
            next_obs = pde_solver.step(obs, action, params)
            replay.add_model_error(field, action, next_obs)

        reward = objective(obs, action, next_obs)
        replay.add(obs, action, reward, next_obs)
        obs = next_obs

policy.update(replay)
```

A PINN can be a world model for planning

Let the PINN approximate a controlled evolution operator

$$\hat{u}_{t+1} = F_\phi(\hat{u}_t, a_t, \eta_t).$$

The policy may be learned by RL or computed online by MPC:

$$\min_{a_{t:t+H-1}} \sum_{k=0}^{H-1} \ell(\hat{u}_{t+k}, a_{t+k}) \quad \text{s.t.} \quad \hat{u}_{t+k+1} = F_\phi(\hat{u}_{t+k}, a_{t+k}).$$

Hybrid advantage

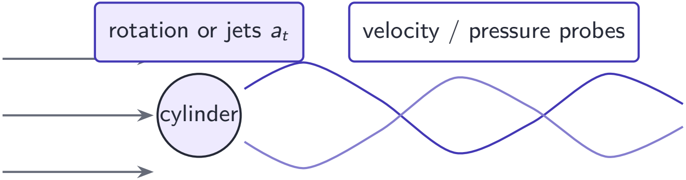
MPC provides explicit horizon and constraints; RL can learn terminal value, residual dynamics, or rapid warm starts. This often gives a clearer safety case than a fully model-free policy.

Motivated by Miyatake et al., PINNs for control of PDE-governed flow systems (2025).

Detailed example: suppressing vortex shedding

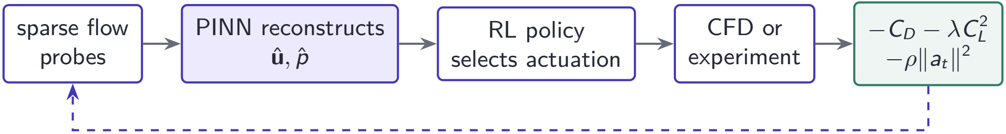
For two-dimensional cylinder flow at Reynolds number $Re = 100$,

$$\nabla \cdot \mathbf{u} = 0, \qquad \frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\nabla p + \frac{1}{Re}\nabla^2 \mathbf{u}.$$



The agent controls cylinder rotation or surface jets to reduce drag and lift fluctuations.

Physics-informed deep RL for flow control



The PINN supplies a physically structured state estimate from limited sensors. The RL policy searches feedback strategies that would be difficult to specify analytically.

Published results are mainly controlled numerical studies. Generalization across Reynolds numbers, geometries, turbulence regimes, and actuator failure remains a substantial research problem.

Motivated by Wang et al., physics-informed DRL for active flow control (2024).

The important PINN–RL failure modes

Failure	Consequence	Required check
Small residual, wrong solution	policy trusts biased state	compare with converged solver / data
Surrogate exploitation	unphysical high reward	uncertainty penalty and truth rollouts
Poor collocation coverage	missed boundary layer or shock	adaptive and held-out residual points
Reward-only physics	approximate constraint violation	projection, barrier, or certified filter
Narrow parameter training	failure under regime change	stress-test geometry and parameters

Physics-informed does not mean physics-correct. The scientific claim depends on external numerical and experimental validation.

Part III: Molecular dynamics and molecular design

RL can choose structures, trajectories, biases, and experiments

Molecular systems contain several different decision problems

Problem	RL decision	Scientific output
Molecular generation	add atom, bond, fragment, or token	candidate structures
Enhanced sampling	choose restart state or bias	conformational ensemble / free energy
Reaction exploration	choose branch or perturbation	transition pathways
Synthesis optimization	choose conditions and measurements	yield, selectivity, robust protocol

These tasks require different dynamics and validation. "RL for molecules" is not one problem.

Why ordinary molecular dynamics is not enough

For coordinates x and momenta p, molecular dynamics evolves

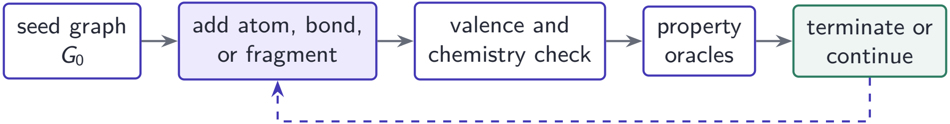
$$\dot{x} = M^{-1}p, \qquad \dot{p} = -\nabla U(x) + \text{thermostat terms.}$$

At equilibrium, configurations follow approximately

$$\rho(x) \propto e^{-\beta U(x)}.$$

Rare transitions may require timescales far beyond feasible simulation. The difficulty is not evaluating one force step; it is discovering relevant slow coordinates and allocating simulation to informative regions without corrupting the target distribution.

Molecular generation as an episodic MDP



- State: molecular graph, remaining step budget, predicted properties.
- Action: chemically permitted edit.
- Reward: terminal or incremental multi-property score.
- Dynamics: deterministic graph edit plus stochastic or uncertain property estimates.

Motivated by Popova, Isayev, and Tropsha, *Deep reinforcement learning for de novo drug design* (2018).

The reward is a molecular specification

A scalar reward may combine

$$R(G) = w_b\, \widehat{A}_{\mathrm{bind}}(G) + w_q\, \mathrm{QED}(G) - w_s\, \mathrm{SA}(G) - w_t\, \widehat{T}_{\mathrm{tox}}(G) - w_u\, \sigma(G),$$

where predicted binding, drug-likeness, synthetic accessibility, toxicity, and epistemic uncertainty compete.

Research issue

A weighted sum hides trade-offs and depends on poorly calibrated predictors. Constrained or Pareto optimization is preferable when one property is non-negotiable.

For example, maximize predicted affinity subject to synthesizability, toxicity, novelty, and uncertainty thresholds.

Detailed example: constrained lead optimization

Start from a validated scaffold and permit a library of medicinal-chemistry edits.

State	graph embedding, attachment sites, assay history, oracle uncertainty
Actions	substitute fragment, change linker, modify ring, terminate
Primary objective	improve measured potency or a cautiously used proxy
Constraints	physicochemical window, alerts, synthesis route, novelty, cost
Feedback	fast predictors $\to$ docking / FEP $\to$ synthesis and assay

The policy should allocate expensive evaluations to candidates for which they can change the decision, not merely maximize a cheap docking score.

Code sketch: a hierarchical molecular evaluation loop

```
graph = seed_scaffold
for step in range(max_edits):
    legal = chemistry.allowed_edits(graph)
    edit = policy.sample(graph, legal)
    candidate = chemistry.apply(graph, edit)

    fast = predictors.evaluate(candidate)
    if fast.toxicity > tox_limit or fast.uncertainty > unc_limit:
        reward, done = -constraint_penalty, True
    else:
        reward = fast.utility - synthesis_cost(candidate)
        if acquisition_value(candidate, fast) > expensive_threshold:
            physics = docking_or_free_energy(candidate)
            reward = update_with_physics(reward, physics)
        done = policy.should_stop(candidate, reward)

    replay.add(graph, edit, reward, candidate, done)
    graph = candidate
    if done: break
```

Laboratory measurements should update both the property model and the policy's uncertainty estimates.

Enhanced sampling changes where computation is spent

Suppose $z = \xi(x)$ is a collective variable and $F(z)$ the free energy. A bias $V_\psi(z)$ can encourage exploration of under-sampled regions:

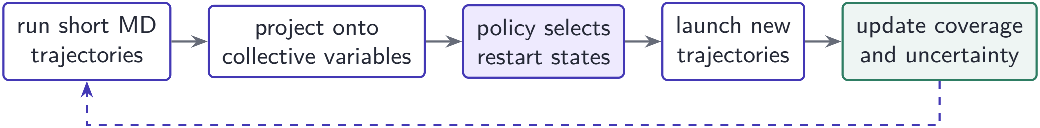
$$U_{\text{biased}}(x) = U(x) + V_\psi(\xi(x)).$$

RL-like formulations can

- select conformations from which new trajectories are launched;
- learn the relative importance of candidate collective variables;
- choose biasing actions to reduce uncertainty in $F(z)$;
- balance exploration of new states with refinement of transition estimates.

Unbiased thermodynamic quantities require valid reweighting or independent sampling; high visitation alone is not a physical result.

Reinforcement-learning-based adaptive sampling



The policy reward can emphasize novelty, uncertainty reduction, or discovery of transitions. The final ensemble must be validated for kinetic and thermodynamic consistency.

Motivated by Shamsi et al., REAP; Hénin et al., *Enhanced sampling methods for molecular dynamics* (2022).

Reinforced dynamics learns an adaptive bias

Reinforced dynamics alternates exploration and estimation:

1. biased MD explores collective-variable space;
2. configurations with high model uncertainty are selected;
3. restrained simulations estimate mean forces;
4. a neural free-energy model and bias are updated.

Interpretation

The method is motivated by reinforcement-learning ideas, but its scientific core is uncertainty-driven adaptive sampling and on-the-fly bias construction. It should not be described as a standard model-free RL agent.

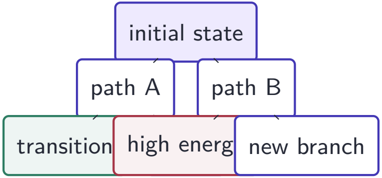
Demonstrations include small peptides and a larger polyalanine system with many collective variables.

Motivated by Zhang, Wang, and E, *Reinforced dynamics for enhanced sampling*.

Reaction pathways can be explored by tree search

Tree-search molecular dynamics treats a molecular conformation as a node and a short MD perturbation as an expansion. Selection can use an upper-confidence rule

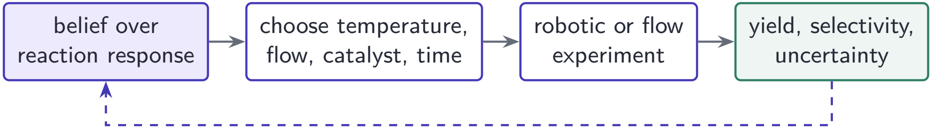
$$a^* = \arg\max_a \left[Q(s,a) + c\sqrt{\frac{\log N(s)}{N(s,a)+1}} \right].$$



The search can accelerate discovery of conformational transitions, but rate estimates require appropriate dynamical treatment rather than raw tree frequencies.

Motivated by Shin et al., tree-search molecular dynamics (2019).

Closed-loop synthesis: RL is useful when experiments have state



If each experiment is independent and the goal is a static optimum, Bayesian optimization is often simpler. RL becomes relevant when actions modify catalyst age, material inventory, process state, or future measurement quality.

A strong research design compares both formulations rather than assuming that a laboratory robot requires RL.

Protein structure prediction is not itself RL

AlphaFold and related structure predictors are essential molecular AI, but their central prediction pipeline is not an RL control problem.

Where RL may contribute

- sequential residue or motif design;
- active selection of expensive assays;
- conformational sampling and docking;
- adaptive experimental protocols.

What must remain separate

- structure prediction accuracy;
- thermodynamic stability;
- biological function;
- synthesizability and safety.

Calling every generative or iterative molecular model “RL” obscures the actual contribution and the relevant validation standard.

Molecular RL needs a validation ladder

chemical validity and uniqueness → predictor + uncertainty → docking, FEP, or quantum chemistry → synthesis and assay → replication and mechanism

- Split by scaffold and time, not only random molecules.
- Detect reward hacking against property predictors.
- Report diversity, constraint satisfaction, and oracle cost.
- Distinguish generated candidates from experimentally validated discoveries.

Motivated by Dodds et al., RL with active learning for molecular design (2024).

Part IV: Robotics and control theory

Learned policies are strongest when embedded in a control architecture

Classical control and RL solve related but different problems

For dynamics $x_{t+1} = f(x_t, u_t, w_t)$, classical optimal control solves

$$\min_{u_{0:T-1}} \sum_{t=0}^{T-1} \ell(x_t, u_t) + \ell_T(x_T) \quad \text{subject to dynamics and constraints.}$$

RL instead learns a policy from interaction or data when f, ℓ, or the relevant state representation is incomplete.

	Classical control	**RL**
Strength	structure, guarantees, data efficiency	nonlinear behaviors, adaptation
Weakness	model/design burden	data, reward, verification burden
Best practice	combine learned components with estimators, MPC, and safety layers	

Detailed example: agile bipedal soccer

DeepMind trained a 20-actuator miniature humanoid for simplified one-versus-one soccer. Individual skills were learned and composed in self-play.

Observation	joint states, inertial sensing, ball and opponent estimates, task phase
Action	target joint positions / low-level motor commands
Reward	ball progress, goals, movement quality, falls, safety-related penalties
Training	simulation, curricula, skill pretraining, self-play, domain variation
Validation	real robot speed, recovery, behavior, and safety interventions

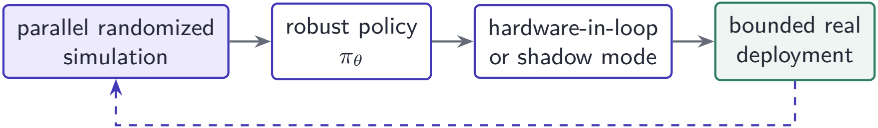
The contribution is not one algorithm. It is a training system that combines task decomposition, simulation, control-frequency decisions, and transfer engineering.

Motivated by Haarnoja et al., *Learning agile soccer skills for a bipedal robot* (Science Robotics, 2024).

The sim-to-real gap is a distribution shift

A policy trained under simulator parameters ξ_{sim} is deployed under unknown real parameters ξ_{real}:

$$P_{\xi_{\text{sim}}}(s' \mid s, a) \neq P_{\xi_{\text{real}}}(s' \mid s, a).$$



Sources of mismatch include contacts, friction, backlash, latency, sensor noise, unmodeled compliance, and visual appearance.

Domain randomization optimizes expected robustness

Sample masses, friction, latency, sensor calibration, terrain, and appearance from $p(\xi)$:

$$\max_{\theta} \mathbb{E}_{\xi \sim p(\xi),\, \tau \sim \pi_\theta, P_\xi} \left[\sum_t \gamma^t r_t \right].$$

Too narrow

- high simulated return;
- brittle real behavior;
- false confidence in the simulator.

Too broad

- unnecessarily conservative policy;
- harder optimization;
- unrealistic parameter combinations.

System identification and real-data calibration should shape $p(\xi)$; randomization is not a substitute for modeling.

Motivated by Muratore et al., *Robot learning from randomized simulations* (2022).

A control barrier function can filter unsafe actions

Let the safe set be $\mathcal{C} = \{x : h(x) \geq 0\}$. For continuous dynamics $\dot{x} = f(x) + g(x)u$, a barrier condition is

$$\nabla h(x)^\top [f(x) + g(x)u] + \alpha(h(x)) \geq 0.$$

The executed action solves

$$u^{\mathrm{safe}} = \arg\min_u \|u - u^{\mathrm{RL}}\|^2 \quad \text{s.t. barrier, torque, and actuator constraints.}$$

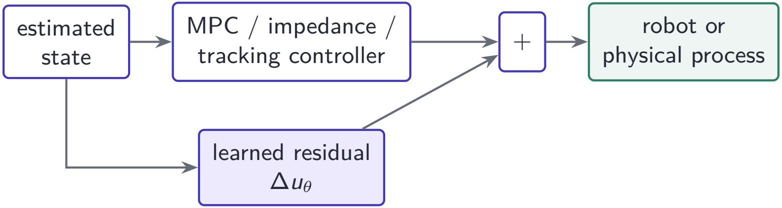
Interpretation

RL proposes a performant action; the safety layer projects it into a locally admissible set. Guarantees depend on model accuracy, state estimation, and feasibility.

Residual RL preserves a trusted controller

Instead of replacing a model-based controller,

$$u_t = u_{\mathrm{base}}(x_t) + \Delta u_\theta(x_t), \qquad \|\Delta u_\theta\| \le \varepsilon.$$



The residual learns friction, contacts, unmodeled aerodynamics, or task-specific corrections while the baseline supplies a stable operating point.

Dexterous manipulation exposes the hardest sim-to-real physics

In-hand manipulation combines many contacts, intermittent friction, tactile uncertainty, occlusion, and high-dimensional actuation.

Successful ingredients

- massive parallel simulation;
- dynamics and visual randomization;
- recurrent policies for hidden state;
- demonstrations or curricula.

Persistent gaps

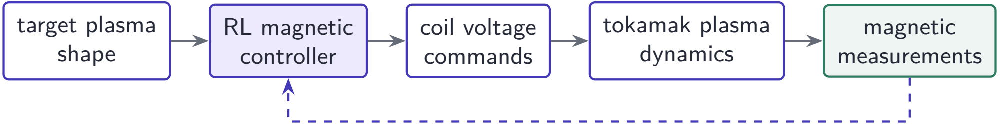
- reliable tactile and contact simulation;
- generalization to new objects and damage;
- sample-efficient real adaptation;
- recoverable failure modes.

Dactyl demonstrated simulation-trained object manipulation on a physical hand, but robust general-purpose dexterity remains open.

Motivated by OpenAI, *Learning dexterity* (2018); recent work on force-sensitive sim-to-real transfer.

A striking control application: shaping tokamak plasma

At the TCV tokamak, a deep RL controller learned in simulation to command magnetic coils and was experimentally deployed for several plasma configurations.



This is an unusually strong demonstration because the claim crosses simulation to real experimental control. It does not imply that an RL policy independently manages an entire fusion reactor.

Motivated by Degrave et al., *Magnetic control of tokamak plasmas through deep RL* (Nature, 2022).

Robotics requires a deployment evidence ladder

randomized simulation → hardware-in-loop and latency → bounded real trials → disturbance and fault testing → operational validation

Report more than mean reward:

- success and failure severity across seeds, hardware units, and environments;
- intervention, reset, collision, wear, energy, and recovery rates;
- comparison with MPC, classical feedback, imitation, and hybrid baselines;
- behavior under sensor failure, latency, payload change, and human proximity.

Part V: Wider research and industrial applications

The maturity of RL differs sharply across domains

Medicine: dynamic treatment regimes from observational data

Sepsis has been formulated as an offline RL problem:

State	vitals, laboratory values, organ function, treatment history
Action	discretized fluid and vasopressor doses
Reward	survival-related outcome and intermediate clinical penalties
Data	retrospective intensive-care records under clinician behavior

Central limitation

The learned policy is evaluated on counterfactual actions that were not taken. Hidden confounding, missingness, action discretization, and limited coverage can make a high estimated value clinically misleading.

Current systems are research decision-support models, not autonomous clinicians. Prospective trials and clinical governance remain necessary.

Motivated by Komorowski et al., *The AI Clinician* (Nature Medicine, 2018); transportability studies through 2025.

Neuroscience: RL is both a model of learning and a controller

RL contributes in two distinct ways.

Computational neuroscience

- reward-prediction-error models of dopamine;
- distributional value coding;
- meta-RL as a theory of adaptive behavior;
- model comparison against neural recordings.

Closed-loop neurotechnology

- state: neural biomarker and symptoms;
- action: stimulation timing or parameters;
- reward: clinical benefit minus side effects;
- constraints: charge, safety, and interpretability.

The computational model may explain behavior without being a biologically literal learning algorithm. Adaptive stimulation requires separate clinical evidence.

Motivated by Muller et al., distributional RL in prefrontal cortex (2024); Hattori et al. (2023).

Finance: execution and market making are sequential control

For optimal execution, a policy chooses how to trade a target inventory over time.

$$r_t = \Delta\mathrm{PnL}_t - \lambda_I q_t^2 - \lambda_C\,\mathrm{cost}_t - \lambda_D\,\mathrm{drawdown}_t.$$

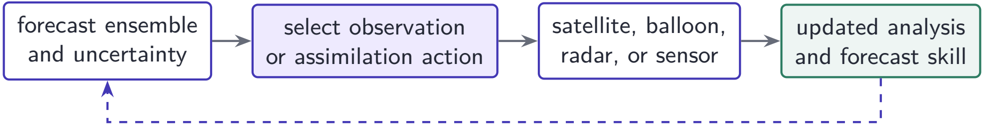
State	order book, inventory q_t, volatility, time remaining
Action	limit price, order size, cancellation, or participation rate
Transition	other strategic agents plus the policy's own market impact
Validation	walk-forward tests, realistic costs, regime and venue shift

Backtests are especially vulnerable to leakage, survivorship bias, nonstationarity, and unrealistic fill simulation. A profitable simulator policy is not evidence of deployable alpha.

Motivated by Nevmyvaka, Feng, and Kearns, RL for optimized trade execution; recent survey of 167 studies (2025).

Weather: prediction is not RL, adaptive observation can be

Graph-based and foundation weather models forecast future atmospheric states; forecasting alone is supervised learning. RL is relevant when the system chooses interventions or observations.



Potential rewards combine forecast-error reduction, observation cost, spatial coverage, and extreme-event value. Chaotic dynamics and rare events make generalization and calibration central.

Motivated by Hammoud et al., deep RL for data assimilation in chaotic systems (2024).

Energy systems: control under renewable uncertainty

RL has been studied for storage dispatch, building control, demand response, voltage control, and network topology actions.

State	load, generation, forecasts, topology, frequency, storage
Action	charge/discharge, switching, curtailment, set-points
Reward	operating cost, losses, emissions, reliability, reserve margin
Hard constraints	power flow, thermal limits, voltage, frequency stability

RL2Grid was designed with transmission-system operators to create more realistic benchmarks, explicitly reflecting combinatorial actions, uncertainty, long horizons, and hard constraints.

Motivated by Marot et al., L2RPN; RL2Grid (2025); Cui et al., Lyapunov-structured frequency control.

Fisheries and aquaculture: feeding is closed-loop control

For precision aquaculture, fish growth and water quality respond to feeding, temperature, oxygen, and stocking conditions.

State	estimated biomass, appetite, temperature, oxygen, ammonia, age
Action	feed rate; in tanks, possibly temperature or aeration set-points
Reward	growth tracking minus feed cost, mortality, and water-quality penalties
Horizon	juvenile stage to target market weight

Q-learning studies have demonstrated growth-trajectory tracking in bioenergetic simulations. The next step is not immediate autonomous feeding: it is validation with uncertain biomass estimation, fish welfare metrics, sensor failures, and farm-specific dynamics.

Motivated by Chahid et al., *Fish growth trajectory tracking via RL* (Aquaculture, 2022).

Wild fisheries: management is a long-horizon POMDP

For a harvested population with uncertain biomass B_t,

$$B_{t+1} = B_t + g(B_t, \omega_t) - H_t + \varepsilon_t,$$

where H_t is harvest and ω_t represents environmental conditions.

- State or belief: stock size, age structure, ecosystem indicators, uncertainty.
- Action: quota, season, spatial closure, monitoring effort.
- Reward: economic value, food supply, stock resilience, ecosystem effects.
- Constraint: probability of stock collapse below an accepted bound.

Model uncertainty and political legitimacy matter more than benchmark return. Policies should be interpretable as management strategies and stress-tested under regime shifts.

Agriculture and ecosystem management

Potential sequential decisions include irrigation, fertilization, greenhouse climate, pest intervention, grazing, invasive-species control, and conservation monitoring.

Why RL may help

- weather and biological delays;
- interacting interventions;
- scarce water and energy;
- spatially distributed systems.

Why it may fail

- simulators omit ecological feedback;
- rewards ignore biodiversity or soil health;
- one season gives few independent episodes;
- policies transfer poorly across sites.

Digital twins, conservative policies, and multi-year field trials are more important than marginal algorithmic gains.

Transport, logistics, and computer networks

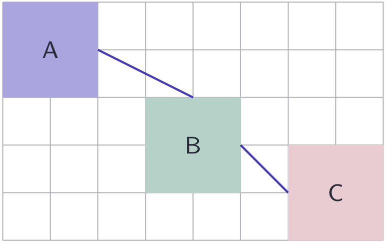
These systems are naturally multi-agent and partially observed.

Application	**Action**	**Research issue**
Traffic signals	phase and duration	network spillback, fairness, incidents
Fleet routing	assign and reposition	demand shift, service equity
Warehouse control	allocate robots and tasks	congestion, deadlines, safety
Congestion control	sending rate / window	competing flows, delayed feedback

Local reward can create global congestion. Centralized training with decentralized execution, graph policies, and mechanism design are active research directions.

Chip placement: impressive result, instructive controversy

Macro placement can be formulated as sequentially placing graph nodes on a grid. Reward combines estimated wirelength, congestion, density, power, and timing.



State: netlist embedding + partial placement

Action: next legal grid location

Reward: proxy for downstream physical design

The 2021 Nature paper reported placements in under six hours comparable or superior to human designs on key metrics. Later work questioned documentation and reproducibility. The case is useful precisely because state-of-the-art claims require accessible baselines and reproducible pipelines.

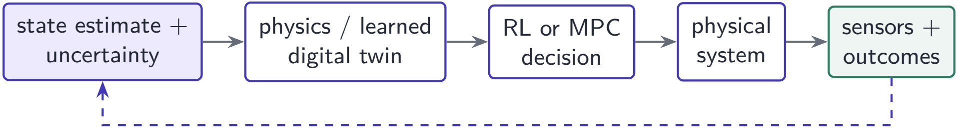
Motivated by Mirhoseini et al., *A graph placement methodology for fast chip design* (Nature, 2021); Cheng et al., reevaluation (2023–2026).

Part VI: What could scientific RL become?

From isolated policies to validated adaptive research systems

Digital twins could connect inference, control, and experiment

A useful digital twin is not a static simulator. It is continuously corrected by observations and used to evaluate interventions.

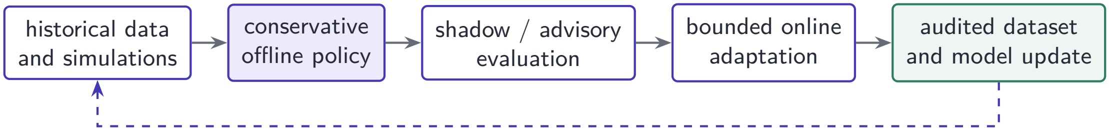


The research frontier is joint uncertainty: the state estimate, dynamics, reward, and constraints may all be wrong. A responsible twin should expose disagreement and trigger information-gathering actions before aggressive control.

Motivated by Reich, digital twins, data assimilation, and optimal control (SIAM, 2025).

Offline-to-online learning can make experiments economical

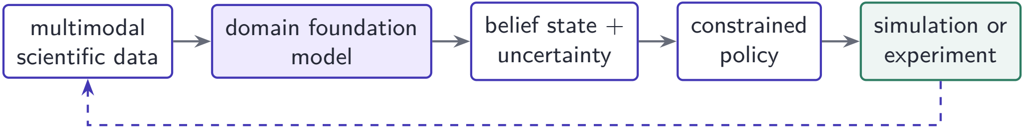
Many domains have abundant imperfect historical data and very limited safe online interaction.



Promising research combines conservative value estimation, uncertainty-aware exploration, human approval, and explicit rollback. The objective is not continuous self-modification; it is controlled acquisition of evidence.

Foundation models could supply scientific state representations

Domain-specific foundation models may encode molecules, proteins, weather fields, robot trajectories, or experimental records. RL can then optimize decisions in the learned representation.



The main risk is representation-induced blindness: the policy can only act on distinctions preserved by the model. Domain validation and out-of-distribution detection remain essential.

Scientific agents could use RL to allocate research effort

An agentic research system may choose among literature search, simulation, analytical derivation, data acquisition, laboratory measurement, and consultation.

State	hypotheses, evidence graph, uncertainty, resources, failed attempts
Action	call simulator, run experiment, query data, revise hypothesis
Reward	information gain and validated progress, not document production
Constraint	safety, budget, attribution, reproducibility, human authorization

What could be transformative

RL may help allocate expensive scientific actions over long horizons. The evaluator must remain external: an agent should not receive reward merely for declaring its own hypothesis confirmed.

The future objective is multi-objective and risk-sensitive

Scientific value cannot usually be reduced to one expected return. A more realistic design is

$$\max_{\pi} \left(J_{\text{knowledge}}, J_{\text{performance}}, -J_{\text{cost}} \right) \quad \text{s.t.} \quad P_{\pi}(\text{critical violation}) \leq \delta.$$

For uncertain model parameters ξ, robust optimization considers

$$\max_{\pi} \min_{\xi \in \Xi} J(\pi; \xi),$$

or a distributionally robust alternative around plausible dynamics.

Research question

Can policies expose a Pareto frontier and allow scientists to choose trade-offs, rather than silently encoding them in reward weights?

Open research problems

Algorithms and models

- reliable offline RL under hidden confounding;
- exploration with hard physical constraints;
- world models with calibrated long-horizon uncertainty;
- continual adaptation without catastrophic drift.

Scientific methodology

- benchmarks that predict real experimental value;
- reproducible simulator and reward specifications;
- causal evaluation of adaptive policies;
- human–policy interfaces for contestable decisions.

The most valuable contributions may be better environments, uncertainty estimates, and validation protocols rather than a new policy optimizer.

A checklist for proposing an RL research project

1. **Sequential need:** What changes because earlier actions were taken?
2. **State:** What is observed, latent, uncertain, and history-dependent?
3. **Action:** Which interventions are genuinely controllable and reversible?
4. **Objective:** Which scientific claim does the reward approximate?
5. **Constraints:** What must never be violated, even during exploration?
6. **Data source:** Simulator, logged data, physical experiment, or hybrid?
7. **Baseline:** Why will MPC, BO, optimal design, or heuristics not suffice?
8. **Validation:** What real evidence would change a domain expert's belief?

Closing perspective

The research opportunity

RL turns scientific models into adaptive decision systems. Its greatest value appears when it combines feedback, physical structure, uncertainty, and carefully chosen experiments.

- PINNs can reconstruct or accelerate dynamics, but external solvers and experiments establish correctness.
- Molecular RL can allocate search and simulation, but assays and thermodynamics establish discovery.
- Robotic RL can synthesize complex behavior, but control structure and hardware testing establish reliability.
- Medicine, finance, weather, energy, and fisheries require domain-specific causal and safety standards.

A credible RL result is not a high return. It is a validated improvement in a real decision process.

Selected sources: physics, molecules, and robotics

- Raissi, Perdikaris, and Karniadakis, *Physics-informed neural networks* (JCP, 2019).
- Wu et al., adaptive sampling for PINNs (CMAME, 2023).
- Farahmand, Nabi, and Nikovski, deep RL for PDE control; Wang et al., physics-informed DRL flow control (2024).
- Popova, Isayev, and Tropsha, deep RL for de novo drug design (2018).
- Zhang, Wang, and E, reinforced dynamics; Hénin et al., enhanced-sampling review.
- Haarnoja et al., agile soccer skills for a bipedal robot (Science Robotics, 2024).
- Degrave et al., magnetic control of tokamak plasmas through deep RL (Nature, 2022).

Selected sources: wider applications and frontiers

- Komorowski et al., *The Artificial Intelligence Clinician learns optimal treatment strategies for sepsis* (Nature Medicine, 2018).
- Muller et al., distributional RL in prefrontal cortex (Nature Neuroscience, 2024).
- Nevmyvaka, Feng, and Kearns, RL for optimized trade execution (ICML, 2006).
- Hammoud et al., RL data assimilation in chaotic systems (2024).
- Marot et al., L2RPN; RL2Grid (2025).
- Chahid et al., fish growth trajectory tracking via RL (Aquaculture, 2022).
- Mirhoseini et al., *A graph placement methodology for fast chip design* (Nature, 2021), with subsequent reproducibility critiques.

Sources represent different evidence levels: simulation, retrospective data, laboratory experiments, and real deployment are not treated as equivalent.